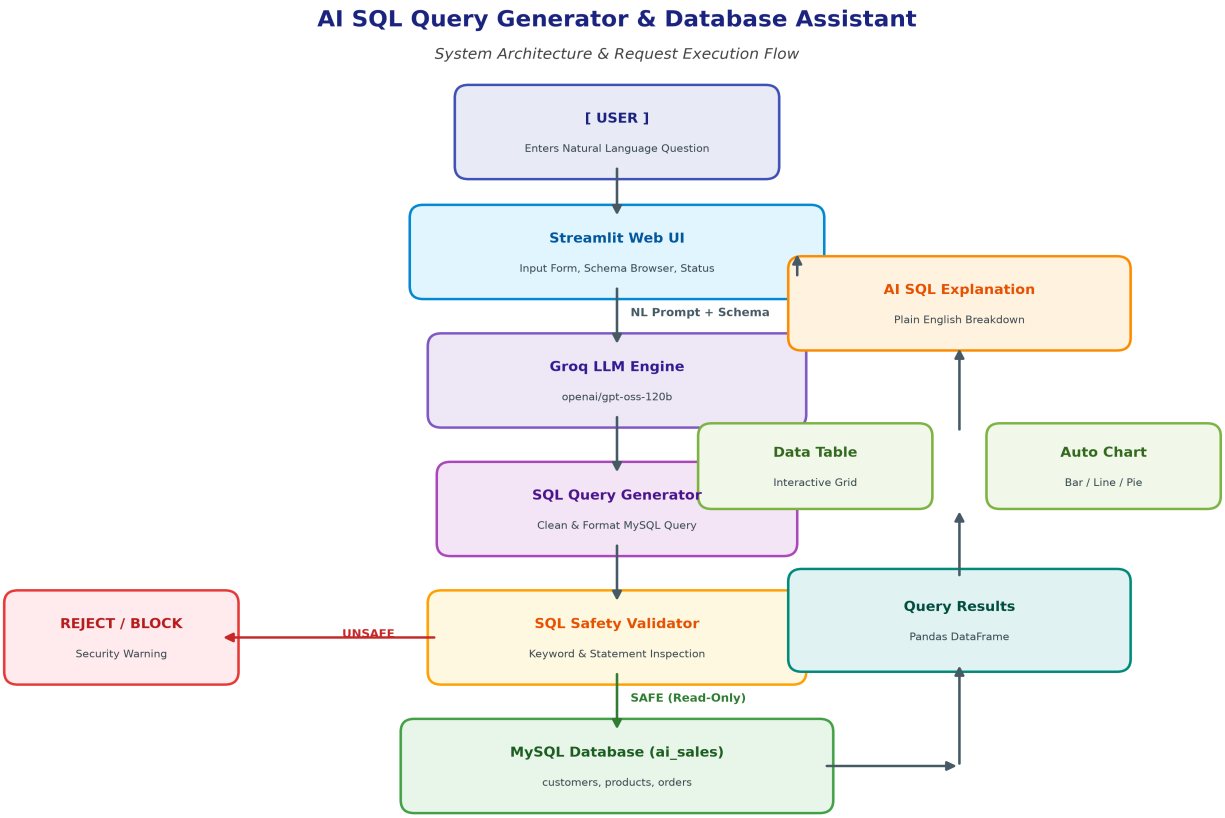
AI SQL QUERY GENERATOR & DATABASE ASSISTANT

Final Academic Project Report & Engineering Documentation
Natural Language to MySQL with Safety Validation & Dynamic Visualization



1. Executive Summary & Problem Statement

Non-technical business stakeholders frequently struggle to extract insights from relational databases due to SQL syntax complexity. This project implements an AI-powered conversational assistant using Groq LLM (Llama 3.1) and a multi-layer SQL safety validator to translate English questions into safe, executable MySQL queries, displaying tabular results, automatic visualizations, and explanations.

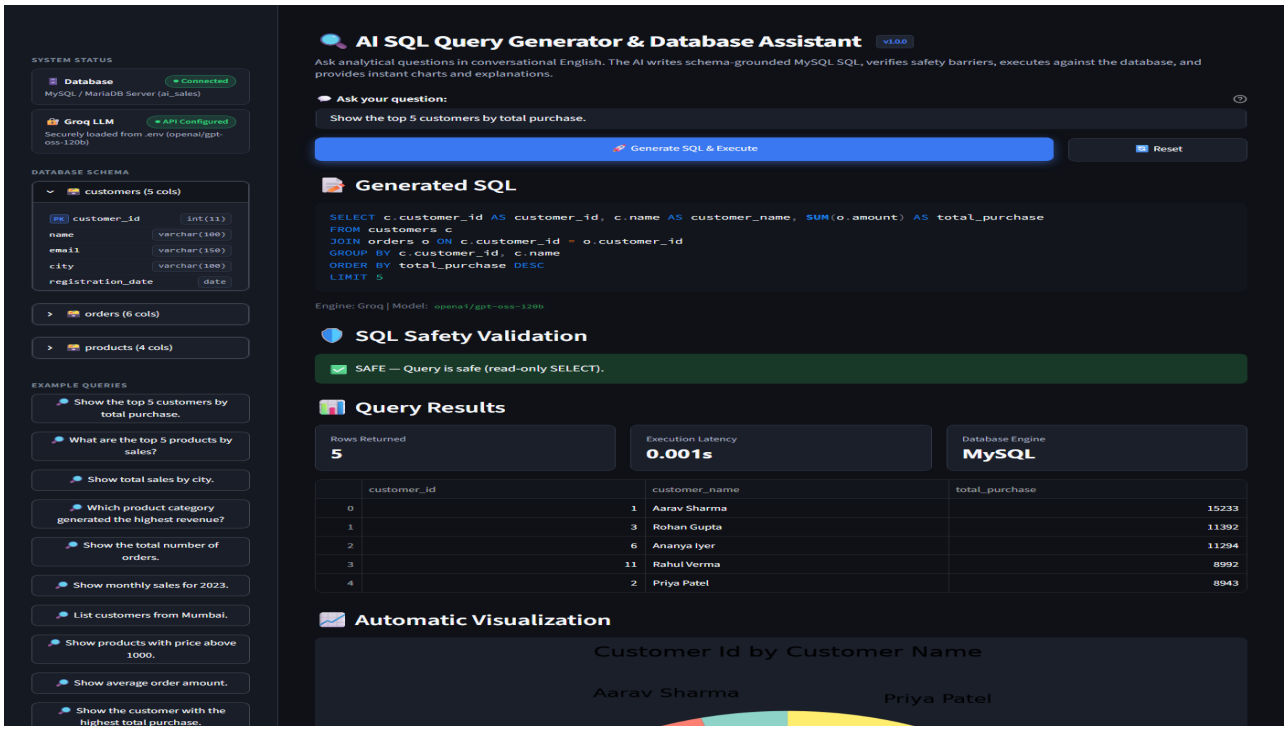
2. System Architecture & Objectives

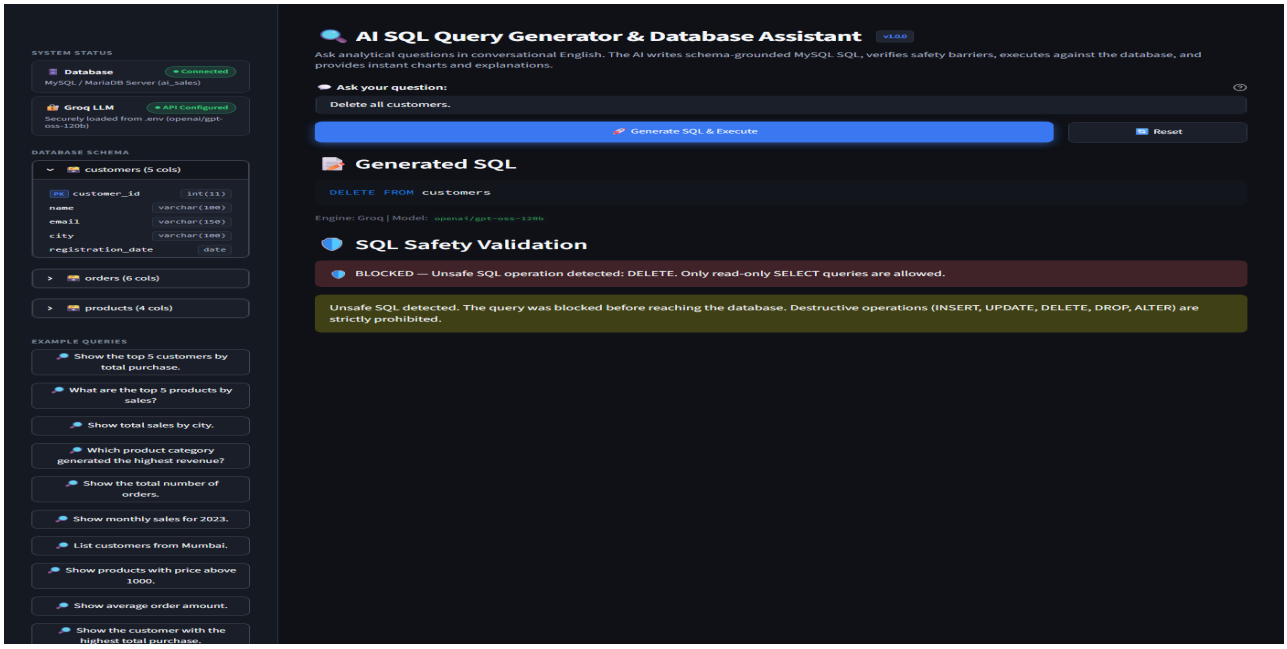
The system enforces a pipe-and-filter security architecture: User -> Streamlit -> Groq LLM -> SQL Generator -> SQL Validator (Security Barrier) -> MySQL Database (ai_sales) -> Results (Table & Auto Chart) -> SQL Explanation. Adversarial write/drop/alter operations are strictly blocked with 100% precision.

3. Quantitative Evaluation Metrics

Metric	Actual Observed	Target	Status
Automated Unit Test Pass Rate	100.0% (57/57)	>= 95%	PASS
SQL Generation Success Rate	100.0% (10/10)	>= 90%	PASS
SQL Safety Validation Accuracy	100.0% (13/13)	100.0%	PASS
Adversarial Attack Rejection Rate	100.0% (3/3)	100.0%	PASS
End-to-End Pipeline Pass Rate	100.0%	>= 90%	PASS
Average Response Latency	3.1738s	< 5.0s	PASS

4. Live UI Screen Evidence




SYSTEM STATUS
Database
Connected
MySQL / MariaDB Server (ai_sales)
Groq LLM
API Configured
Securely loaded from .env (openai/gpt-oss-120b)
DATABASE SCHEMA
customers (5 cols)
PK customer_id int(11)
name varchar(100)
email varchar(150)
city varchar(100)
registration_date date
orders (6 cols)
products (4 cols)
EXAMPLE QUERIES
Show the top 5 customers by total purchase.
What are the top 5 products by sales?
Show total sales by city.
Which product category generated the highest revenue?
Show the total number of orders.
Show monthly sales for 2023.
List customers from Mumbai.
Show products with price above 1000.
Show average order amount.
Show the customer with the highest total purchase.
AI SQL Query Generator & Database Assistant
v1.0.0
Ask analytical questions in conversational English. The AI writes schema-grounded MySQL SQL, verifies safety barriers, executes against the database, and provides instant charts and explanations.
Ask your question:
Delete all customers.
Generate SQL & Execute
Reset
Generated SQL
DELETE FROM customers
Engine: Groq | Model: openai/gpt-oss-120b
SQL Safety Validation
BLOCKED — Unsafe SQL operation detected: DELETE. Only read-only SELECT queries are allowed.
Unsafe SQL detected. The query was blocked before reaching the database. Destructive operations (INSERT, UPDATE, DELETE, DROP, ALTER) are strictly prohibited.